

Statistics for Data Science - 2

Week 8 Graded Assignment Solution

1. Let $X_1, X_2, \dots, X_n$ be i.i.d. samples from a distribution X with mean μ and standard deviation σ . Let $\hat{\mu} = 6 \left(\frac{X_1 + X_2 + \dots + X_n}{n} \right)$ be an estimator of μ .

i) Is the estimator unbiased?

a) Yes

b) No

Solution:

$$\begin{aligned} E[\hat{\mu}] &= E \left[6 \left(\frac{X_1 + X_2 + \dots + X_n}{n} \right) \right] \\ &= \frac{6}{n} (n\mu) \\ &= 6\mu \end{aligned}$$

And

$$\text{Bias}(\hat{\mu}, \mu) = E[\hat{\mu}] - \mu = 6\mu - \mu = 5\mu$$

Since, $\text{Bias}(\hat{\mu}, \mu) \neq 0$, therefore the estimator is not unbiased.

ii) Find the risk of $\hat{\mu}$.

- (a) $\frac{36\sigma^2}{n} + 25\mu^2$
- (b) $\frac{36\sigma^2}{n} + 5\mu$
- (c) $\frac{6\sigma^2}{n} + 25\mu^2$
- (d) $\frac{6\sigma^2}{n} + 5\mu$

Solution:

$$\begin{aligned} \text{Var}(\hat{\mu}) &= \text{Var} \left[6 \left(\frac{X_1 + X_2 + \dots + X_n}{n} \right) \right] \\ &= \frac{36}{n^2} (n\sigma^2) \\ &= \frac{36\sigma^2}{n} \end{aligned}$$

$$\begin{aligned}
\text{Risk}(\hat{\mu}) &= \text{Bias}(\hat{\mu}, \mu)^2 + \text{Var}(\hat{\mu}) \\
&= (5\mu)^2 + \frac{36\sigma^2}{n} \\
&= 25\mu^2 + \frac{36\sigma^2}{n}
\end{aligned}$$

2. Consider a sample of iid random variables $X_1, X_2, \dots, X_n$, where $n > 20, E[X_i] = \mu, \text{Var}(X_i) = \sigma^2$ and the estimator of $\mu, \hat{\mu}_n = \frac{1}{n-20} \sum_{i=21}^n X_i$. Find the MSE of $\hat{\mu}_n$.

- a) $\frac{\sigma}{n-20}$
- b) $\frac{\sigma^2}{n-20}$
- c) $\frac{\sigma^2}{n-21}$
- d) $\frac{\sigma}{n}$

Solution:

$$\begin{aligned}
E[\hat{\mu}_n] &= E\left[\frac{1}{n-20} \sum_{i=21}^n X_i\right] \\
&= \frac{(n-20)\mu}{n-20} \\
&= \mu
\end{aligned}$$

This implies that

$$\text{Bias}(\hat{\mu}_n, \mu) = E[\hat{\mu}_n] - \mu = \mu - \mu = 0$$

$$\begin{aligned}
\text{Var}(\hat{\mu}_n) &= \text{Var}\left[\frac{1}{n-20} \sum_{i=21}^n X_i\right] \\
&= \frac{1}{(n-20)^2} \sum_{i=21}^n \text{Var}(X_i) \\
&= \frac{1}{(n-20)^2} [(n-20)\sigma^2] \\
&= \frac{\sigma^2}{(n-20)}
\end{aligned}$$

$$\begin{aligned}
\text{Risk}(\hat{\mu}) &= \text{Bias}(\hat{\mu}, \mu)^2 + \text{Var}(\hat{\mu}) \\
&= 0 + \frac{\sigma^2}{(n-20)} \\
&= \frac{\sigma^2}{(n-20)}
\end{aligned}$$

3. Let $X_1, X_2, \dots, X_n \sim \text{iid } X$, where X is a random variable with density function

$$f_X(x) = \begin{cases} \frac{\theta}{x^{\theta+1}}, & x > 1, \\ 0, & \text{otherwise.} \end{cases}$$

The mean of the random variable X is $\frac{\theta}{\theta-1}$. Find an estimator of θ using method of moments.

- (a) $\frac{X_1 + X_2 + \dots + X_n}{X_1 + X_2 + \dots + X_n - 1}$
 (b) $\frac{X_1 + X_2 + \dots + X_n}{1 - X_1 + X_2 + \dots + X_n}$
 (c) $\frac{X_1 + X_2 + \dots + X_n}{X_1 + X_2 + \dots + X_n - n}$
 (d) $\frac{X_1 + X_2 + \dots + X_n}{n - X_1 + X_2 + \dots + X_n}$

Solution: The mean of the random variable X is $\frac{\theta}{\theta-1}$.
 So,

$$\begin{aligned} M_1 &= \frac{\theta}{\theta-1} \\ \Rightarrow M_1\theta - M_1 &= \theta \\ \Rightarrow \theta &= \frac{M_1}{M_1 - 1} \\ \Rightarrow \theta &= \frac{\frac{X_1 + X_2 + \dots + X_n}{n}}{\frac{X_1 + X_2 + \dots + X_n}{n} - 1} \\ \Rightarrow \theta &= \frac{X_1 + X_2 + \dots + X_n}{X_1 + X_2 + \dots + X_n - n} \end{aligned}$$

Therefore the estimator of θ is $\frac{X_1 + X_2 + \dots + X_n}{X_1 + X_2 + \dots + X_n - n}$.

4. Let $X_1, X_2, X_3 \sim \text{iid Binomial}(4, \theta)$. Given a random sample $(1, 4, 2)$, find the maximum likelihood estimate of θ .
- a) $\frac{2}{3}$
 b) $\frac{7}{12}$
 c) $\frac{1}{3}$
 d) $\frac{5}{12}$

Solution: $X_i \sim \text{Binomial}(4, \theta)$
 $\Rightarrow f_{X_i}(x) = {}^4C_x \theta^x (1 - \theta)^{4-x}$

Likelihood function is given by

$$L(x_1, x_2, x_3) = \prod_{i=1}^3 f_{X_i}(x_i)$$

$$\Rightarrow L(x_1, x_2, x_3) = {}^4C_{x_1} \theta^{x_1} (1 - \theta)^{4-x_1} \times {}^4C_{x_2} \theta^{x_2} (1 - \theta)^{4-x_2} \times {}^4C_{x_3} \theta^{x_3} (1 - \theta)^{4-x_3}$$

$$L(1, 4, 2) = {}^4C_1 {}^4C_4 {}^4C_2 \theta^{(1+4+2)} (1 - \theta)^{12-(1+4+2)}$$

$$= 24\theta^7 (1 - \theta)^5$$

$$\Rightarrow \log(L(1, 4, 2)) = \log(24) + 7\log(\theta) + 5\log(1 - \theta)$$

Therefore, ML estimator for θ is given by

$$\hat{\theta} = \arg \max_{\theta} [\log(24) + 7\log(\theta) + 5\log(1 - \theta)]$$

$$\text{Let } Y = \log(24) + 7\log(\theta) + 5\log(1 - \theta)$$

$$\Rightarrow \frac{dY}{d\theta} = \frac{7}{\theta} - \frac{5}{1 - \theta}$$

Now we will equate this value to zero and find the value of θ

$$\frac{7}{\theta} - \frac{5}{1 - \theta} = 0 \Rightarrow \theta = \frac{7}{12}$$

$$\Rightarrow \hat{\theta}_{ML} = \frac{7}{12}$$

5. Let $X_1, X_2, \dots, X_n \sim \text{iid } X$, where X is a random variable with density function

$$f_X(x) = \begin{cases} e^{-(x-\theta)}, & x > \theta, \\ 0, & \text{otherwise.} \end{cases}$$

i) The mean of the distribution is $\theta + 1$. Find the estimator of θ using method of moments. [1 mark]

(a) $\frac{X_1 + X_2 + \dots + X_n}{n}$

(b) $\frac{X_1 + X_2 + \dots + X_n - n}{n}$

(c) $\frac{X_1 + X_2 + \dots + X_n - n}{n}$

(d) $\frac{1}{n - X_1 + X_2 + \dots + X_n}$

Solution: The mean of the random variable X is $\theta + 1$.
So,

$$\begin{aligned} M_1 &= \theta + 1 \\ \Rightarrow \theta &= M_1 - 1 \\ \Rightarrow \theta &= \frac{X_1 + X_2 + \dots + X_n}{n} - 1 \\ \Rightarrow \theta &= \frac{X_1 + X_2 + \dots + X_n - n}{n} \end{aligned}$$

Therefore the estimator of θ is $\frac{X_1 + X_2 + \dots + X_n - n}{n}$.

ii) Is the method of moments estimator unbiased?

a) Yes

b) No

Solution:

Estimator of θ is

$$\hat{\theta} = \frac{X_1 + X_2 + \dots + X_n - n}{n}$$

$$\begin{aligned} E[\hat{\theta}] &= E\left[\frac{X_1 + X_2 + \dots + X_n - n}{n}\right] \\ &= \frac{1}{n}(E[X_1] + E[X_2] + \dots + E[X_n] - n) \\ &= \frac{1}{n}(n\theta + n - n) \\ &= \theta \end{aligned}$$

And

$$\text{Bias}(\hat{\theta}, \theta) = E[\hat{\theta}] - \theta = \theta - \theta = 0$$

Since, $\text{Bias}(\hat{\theta}, \theta) = 0$, therefore the estimator is unbiased.

6. Suppose it is known that a sample consisting of the values 10, 12, 15, 16.5, 18, 19, 20 and 21.5 comes from a population with the density function

$$f(x) = \begin{cases} \frac{1}{\theta} e^{-\frac{x}{\theta}}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find the maximum likelihood estimate of θ . Enter your answer correct to one decimal.

Solution:

$$\begin{aligned}
 L(x_1, x_2, \dots, x_n) &= \prod_{i=1}^n f_X(x_i) \\
 &= \prod_{i=1}^n \frac{1}{\theta} e^{\frac{-x_i}{\theta}} \\
 &= \frac{1}{\theta^n} \left(e^{\frac{-x_1}{\theta}} e^{\frac{-x_2}{\theta}} \dots e^{\frac{-x_n}{\theta}} \right) \\
 &= \frac{1}{\theta^n} \left(e^{\frac{-(x_1+x_2+\dots+x_n)}{\theta}} \right)
 \end{aligned}$$

$$\Rightarrow \log(L(x_1, x_2, \dots, x_n)) = -n \log(\theta) - \frac{(x_1 + x_2 + \dots + x_n)}{\theta}$$

Therefore, ML estimator for θ is given by

$$\hat{\theta} = \arg \max_{\theta} \left[-n \log(\theta) - \frac{(x_1 + x_2 + \dots + x_n)}{\theta} \right]$$

$$\begin{aligned}
 \text{Let } Y &= -n \log(\theta) - \frac{(x_1 + x_2 + \dots + x_n)}{\theta} \\
 \Rightarrow \frac{dY}{d\theta} &= -\frac{n}{\theta} + \frac{(x_1 + x_2 + \dots + x_n)}{\theta^2}
 \end{aligned}$$

Now we will equate this value to zero and find the value of θ .

$$\begin{aligned}
 \Rightarrow -\frac{n}{\theta} + \frac{(x_1 + x_2 + \dots + x_n)}{\theta^2} &= 0 \\
 \Rightarrow \theta &= \frac{x_1 + x_2 + \dots + x_n}{n} \\
 \Rightarrow \hat{\theta} &= \frac{x_1 + x_2 + \dots + x_n}{n}
 \end{aligned}$$

Therefore, maximum likelihood estimate of θ for the given sample will be

$$\begin{aligned}
 \hat{\theta} &= \frac{10 + 12 + 15 + 16.5 + 18 + 19 + 20 + 21.5}{8} \\
 &= \frac{132}{8} \\
 &= 16.5
 \end{aligned}$$

7. Let X be a discrete random variable with the following probability mass function

x	1	2	3	4
$f_X(x)$	$\frac{1-p}{2}$	$\frac{p}{2}$	$\frac{1-p}{2}$	$\frac{p}{2}$

Table 8.1.G: PMF of X

Suppose a sample consisting of the values 2, 2, 4, 3, 1, 3, 1 and 2 is taken from the random variable X . Find the estimate of p using method of moments. Enter your answer correct to two decimals accuracy.

Solution:

$$\begin{aligned} E[X] &= 1 \times \frac{1-p}{2} + 2 \times \frac{p}{2} + 3 \times \frac{1-p}{2} + 4 \times \frac{p}{2} \\ &= \frac{(1-p) + 2p + 3(1-p) + 4p}{2} \\ &= p + 2 \end{aligned}$$

Now

$$M_1 = E[X] = p + 2$$

$$\Rightarrow p = M_1 - 2$$

Therefore, estimate of p will be

$$\frac{X_1 + X_2 + \dots + X_n}{n} - 2.$$

So, the estimate of p for the given sample will be

$$\begin{aligned} \hat{p} &= \frac{2 + 2 + 4 + 3 + 1 + 3 + 1 + 2}{8} - 2 \\ &= \frac{18}{8} - 2 \\ &= 0.25 \end{aligned}$$

Use the following values of CDF of standard normal distribution to answer the questions:

$$F_Z(1.64) = 0.90, F_Z(1.96) = 0.95$$

8. The weights (in grams) of mangoes grown in a certain area are normally distributed with mean μ and standard deviation 40. The weights from a random sample of mangoes are as follows:

220, 210, 240, 260, 235, 225, 270, 300, 200.

Find a 95% confidence interval for the mean weight of mangoes.

- a) [203.87, 256.13]
- b) [213.87, 266.13]
- c) [230, 280]
- d) [215.13, 235.87]

Solution:

$n = 9$, $\hat{\mu} = 240$ and $\sigma = 40$.

$\beta = 0.95$, using CDF of Normal(0, 1),

$$\frac{\alpha}{\sigma/\sqrt{n}} = 1.96$$

$$\alpha = 1.96 \times \frac{40}{\sqrt{9}} = 26.13$$

$$P(|\hat{\mu} - \mu| < 26.13) = 0.95$$

So, 95% confidence interval is $[240 - 26.13, 240 + 26.13]$ i.e. $[213.87, 266.13]$

9. From past experience it is known that the weights of seer fish grown at a commercial hatchery are normal with a mean that varies from season to season but with a standard deviation that remains fixed at 0.2 kilogram. If we want to be 90% certain that our estimate of the present season's mean weight of a seer fish is correct to within 0.01 kilograms, how large a sample is needed?

Solution:

Let X denote the weights of seer fish.

Given that $\sigma = 0.2$

To find the value of n such that $P(|\hat{\mu} - \mu| \leq 0.01) = 0.90$

$$\begin{aligned} P(|\hat{\mu} - \mu| \leq 0.01) &= 0.90 \\ \Rightarrow P\left(\left|\frac{\hat{\mu} - \mu}{\sigma/\sqrt{n}}\right| \leq \frac{0.01}{\sigma/\sqrt{n}}\right) &= 0.90 \\ \Rightarrow P\left(|Z| \leq \frac{0.01}{\sigma/\sqrt{n}}\right) &= 0.90 \end{aligned}$$

$$\frac{0.01}{\sigma/\sqrt{n}} = 1.64$$

$$\Rightarrow \sqrt{n} = 0.2 \times \frac{1.64}{0.01}$$

$$\Rightarrow n = 1075.84$$

Therefore the sample size should be 1076.

10. The distribution of heights of a certain population of women is normally distributed with μ unknown and σ unknown. We observe a random sample (in centimeters): 160, 155, 168, 167, 162, 150, 152, 148, 164.
Find a 95% confidence interval for μ . Use $P(-2.30 < T_8 < 2.30) = 0.95$ where T_8 is t -distribution with degree of freedom 8.

a) [152.73, 164.15]

- b) [156.67, 160.2]
- c) [160.28, 167.72]
- d) [150.34, 165.66]

Solution:

$n = 9, \hat{\mu} = 158.44$ and $S^2 = 55.52 \Rightarrow S = 7.45$

Using t -distribution, $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$

$$\begin{aligned}\frac{\alpha}{S/\sqrt{n}} &= 2.30 \\ \alpha &= 2.30 \times \frac{7.45}{\sqrt{9}} = 5.71 \\ P(|\hat{\mu} - \mu| < 5.71) &= 0.95\end{aligned}$$

So, 95% confidence interval is $[158.44 - 5.71, 158.44 + 5.71]$ i.e. $[152.73, 164.15]$.